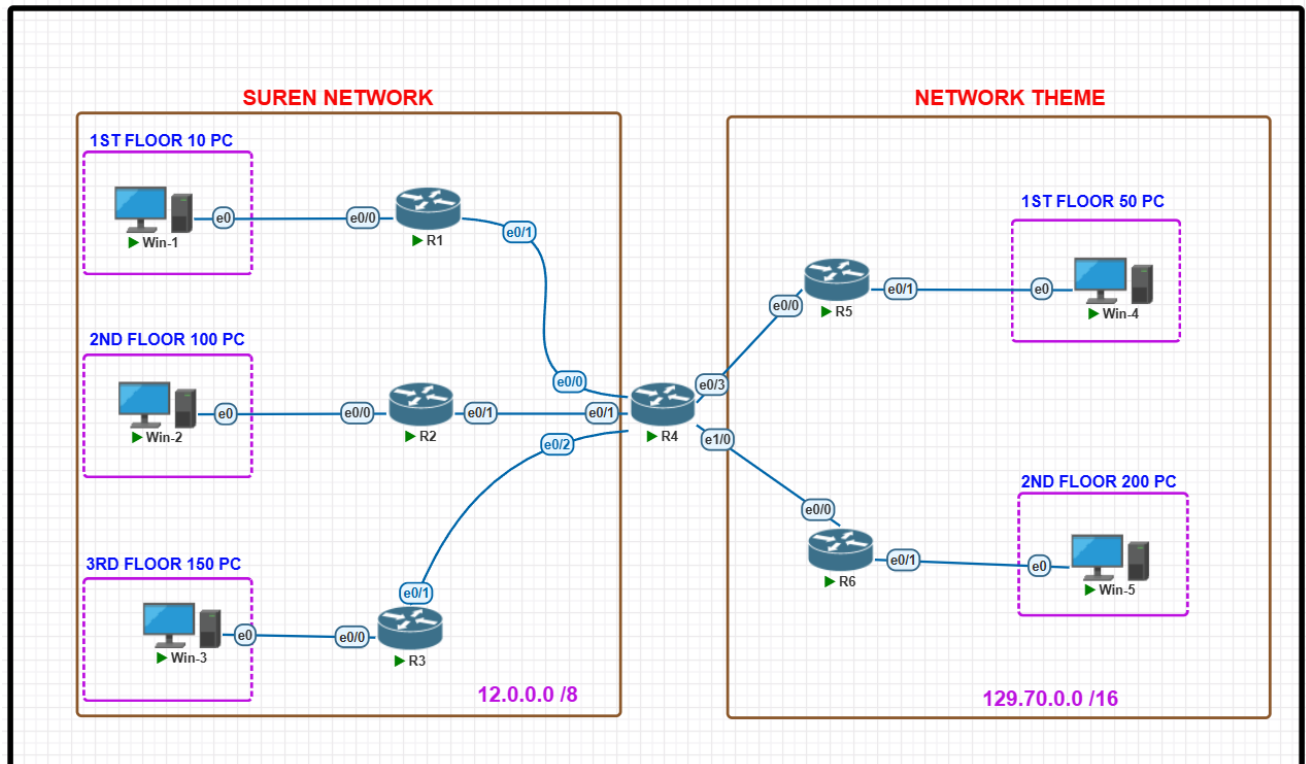


Subnet Project



Pool Configuration in R4

```
Router#sh ip dhcp pool

Pool SN 1 :
  Utilization mark (high/low) : 100 / 0
  Subnet size (first/next) : 0 / 0
  Total addresses : 14
  Leased addresses : 1
  Pending event : none
  1 subnet is currently in the pool :
  Current index IP address range Leased addresses
  50.0.1.131 50.0.1.129 - 50.0.1.142 1

Pool SN 2 :
  Utilization mark (high/low) : 100 / 0
  Subnet size (first/next) : 0 / 0
  Total addresses : 126
  Leased addresses : 1
  Pending event : none
  1 subnet is currently in the pool :
  Current index IP address range Leased addresses
  50.0.1.3 50.0.1.1 - 50.0.1.126 1

Pool SN 3 :
  Utilization mark (high/low) : 100 / 0
  Subnet size (first/next) : 0 / 0
  Total addresses : 254
  Leased addresses : 1
  Pending event : none
  1 subnet is currently in the pool :
  Current index IP address range Leased addresses
  50.0.0.3 50.0.0.1 - 50.0.0.254 1

Pool NT 1 :
  Utilization mark (high/low) : 100 / 0
  Subnet size (first/next) : 0 / 0
  Total addresses : 62
  Leased addresses : 1
  Pending event : none
  1 subnet is currently in the pool :
  Current index IP address range Leased addresses
  129.70.1.42 129.70.1.1 - 129.70.1.62 1

Pool NT 2 :
  Utilization mark (high/low) : 100 / 0
  Subnet size (first/next) : 0 / 0
  Total addresses : 254
  Leased addresses : 1
  Pending event : none
```

Route and IP in R4

```
R4
Interface      IP-Address      OK? Method Status      Prot
Ethernet0/0    50.0.1.154      YES manual up          up
Ethernet0/1    50.0.1.150      YES manual up          up
Ethernet0/2    50.0.1.146      YES manual up          up
Ethernet0/3    129.70.1.66     YES manual up          up
Ethernet1/0    129.70.1.70     YES manual up          up
Ethernet1/1    unassigned      YES unset  administratively down down
Ethernet1/2    unassigned      YES unset  administratively down down
Ethernet1/3    unassigned      YES unset  administratively down down

Router#sh ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       I - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
       a - application route
       + - replicated route, % - next hop override, p - overrides from PfR

Gateway of last resort is not set

50.0.0.0/8 is variably subnetted, 9 subnets, 5 masks
S    50.0.0.0/24 [1/0] via 50.0.1.145
S    50.0.1.0/25 [1/0] via 50.0.1.149
S    50.0.1.128/28 [1/0] via 50.0.1.153
C    50.0.1.144/30 is directly connected, Ethernet0/2
L    50.0.1.146/32 is directly connected, Ethernet0/2
C    50.0.1.148/30 is directly connected, Ethernet0/1
L    50.0.1.150/32 is directly connected, Ethernet0/1
C    50.0.1.152/30 is directly connected, Ethernet0/0
L    50.0.1.154/32 is directly connected, Ethernet0/0
129.70.0.0/16 is variably subnetted, 6 subnets, 4 masks
S    129.70.0.0/24 [1/0] via 129.70.1.69
S    129.70.1.0/26 [1/0] via 129.70.1.65
C    129.70.1.64/30 is directly connected, Ethernet0/3
L    129.70.1.66/32 is directly connected, Ethernet0/3
C    129.70.1.68/30 is directly connected, Ethernet1/0
L    129.70.1.70/32 is directly connected, Ethernet1/0
Router#
```

All other Router Configuration

```
R1
R1(config-if)#
R1#sh
*Aug 4 08:49:13.794: %SYS-5-CONFIG_I: Configured from console by console
R1#sh ip int brief
Interface      IP-Address      OK? Method Status      Prot
Ethernet0/0    50.0.1.129      YES manual up          up
Ethernet0/1    50.0.1.153      YES manual up          up
Ethernet0/2    unassigned      YES TFTP  down        down
Ethernet0/3    unassigned      YES TFTP  down        down

R1#sh ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       I - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
       a - application route
       + - replicated route, % - next hop override, p - overrides from PfR

Gateway of last resort is 50.0.1.154 to network 0.0.0.0

S*   0.0.0.0/0 [1/0] via 50.0.1.154
50.0.0.0/8 is variably subnetted, 4 subnets, 3 masks
C    50.0.1.128/28 is directly connected, Ethernet0/0
L    50.0.1.129/32 is directly connected, Ethernet0/0
C    50.0.1.152/30 is directly connected, Ethernet0/1
L    50.0.1.153/32 is directly connected, Ethernet0/1
R1#sh ip helper-address
Interface      Helper-Address  VPM VRG Name      VRG State
Ethernet0/0    50.0.1.154     0   None           Unknown
R1#
```

```
R2
R2(config-if)#
R2#sh
*Aug 4 09:02:44.926: %SYS-5-CONFIG_I: Configured from console by console
R2#sh ip int brief
Interface IP-Address OK? Method Status Prot
Gig0/0 50.0.1.1 YES manual up
Ethernet0/0 50.0.1.149 YES manual up up
Ethernet0/2 unassigned YES TFTP down down
Ethernet0/3 unassigned YES TFTP down down

R2#sh ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
E1 - OSPF external type 1, E2 - OSPF external type 2
I - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
ia - IS-IS inter area, * - candidate default, U - per-user static route
o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
a - application route
+ - replicated route, % - next hop override, p - overrides from PFR

Gateway of last resort is 50.0.1.150 to network 0.0.0.0

S* 0.0.0.0/0 [1/0] via 50.0.1.150
50.0.0.0/8 is variably subnetted, 4 subnets, 3 masks
C 50.0.1.0/25 is directly connected, Ethernet0/0
L 50.0.1.1/32 is directly connected, Ethernet0/0
C 50.0.1.149/30 is directly connected, Ethernet0/1
L 50.0.1.149/32 is directly connected, Ethernet0/1
R2#sh ip helper-address
Interface Helper-Address VRF Name VRF State
Ethernet0/0 50.0.1.150 0 None Unknown
R2#
```

```
R3
R3(config-if)#
R3#sh
*Aug 4 09:03:28.729: %SYS-5-CONFIG_I: Configured from console by console
R3#sh ip int brief
Interface IP-Address OK? Method Status Prot
Gig0/0 50.0.0.1 YES manual up
Ethernet0/0 50.0.1.145 YES manual up up
Ethernet0/2 unassigned YES TFTP down down
Ethernet0/3 unassigned YES TFTP down down

R3#sh ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
E1 - OSPF external type 1, E2 - OSPF external type 2
I - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
ia - IS-IS inter area, * - candidate default, U - per-user static route
o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
a - application route
+ - replicated route, % - next hop override, p - overrides from PFR

Gateway of last resort is 50.0.1.146 to network 0.0.0.0

S* 0.0.0.0/0 [1/0] via 50.0.1.146
50.0.0.0/8 is variably subnetted, 4 subnets, 3 masks
C 50.0.0.0/24 is directly connected, Ethernet0/0
L 50.0.0.1/32 is directly connected, Ethernet0/0
C 50.0.1.144/30 is directly connected, Ethernet0/1
L 50.0.1.145/32 is directly connected, Ethernet0/1
R3#sh ip helper-address
Interface Helper-Address VRF Name VRF State
Ethernet0/0 50.0.1.146 0 None Unknown
R3#
```

```
R5
R5#sh ip int brief
Interface      IP-Address      OK? Method Status      Prot
Ethernet0/0    129.70.1.65     YES manual up          up
Ethernet0/1    129.70.1.1      YES manual up          up
Ethernet0/2    unassigned      YES TFTP  down        down
Ethernet0/3    unassigned      YES TFTP  down        down

R5#sh ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
       a - application route
       + - replicated route, % - next hop override, p - overrides from PFR

Gateway of last resort is 129.70.1.66 to network 0.0.0.0

S* 0.0.0.0/0 [1/0] via 129.70.1.66
   129.70.0.0/16 is variably subnetted, 4 subnets, 3 masks
C    129.70.1.0/26 is directly connected, Ethernet0/1
L    129.70.1.1/32 is directly connected, Ethernet0/1
C    129.70.1.64/30 is directly connected, Ethernet0/0
L    129.70.1.65/32 is directly connected, Ethernet0/0

R5#sh ip helper-address
Interface      Helper-Address  VFN VRG Name      VRG State
Ethernet0/1    129.70.1.66    0   None            Unknown
R5
```

```
R6
R6#sh ip int brief
Interface      IP-Address      OK? Method Status      Prot
Ethernet0/0    129.70.1.69     YES manual up          up
Ethernet0/1    129.70.0.1      YES manual up          up
Ethernet0/2    unassigned      YES TFTP  down        down
Ethernet0/3    unassigned      YES TFTP  down        down

R6#sh ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
       a - application route
       + - replicated route, % - next hop override, p - overrides from PFR

Gateway of last resort is 129.70.1.70 to network 0.0.0.0

S* 0.0.0.0/0 [1/0] via 129.70.1.70
   129.70.0.0/16 is variably subnetted, 4 subnets, 3 masks
C    129.70.0.0/24 is directly connected, Ethernet0/1
L    129.70.0.1/32 is directly connected, Ethernet0/1
C    129.70.1.68/30 is directly connected, Ethernet0/0
L    129.70.1.69/32 is directly connected, Ethernet0/0

R6#sh ip helper-address
Interface      Helper-Address  VFN VRG Name      VRG State
Ethernet0/1    129.70.1.70    0   None            Unknown
R6
```

IP DHCP Binding in R4

```
R4
Pending event      : none
1 subnet is currently in the pool :
Current index      IP address range      Leased addresses
50.0.1.3           50.0.1.1 - 50.0.1.126              1

Pool SM 3 :
Utilization mark (high/low) : 100 / 0
Subnet size (first/next)    : 0 / 0
Total addresses              : 254
Leased addresses             : 1
Pending event                : none
1 subnet is currently in the pool :
Current index      IP address range      Leased addresses
50.0.0.3           50.0.0.1 - 50.0.0.254    1

Pool NT 1 :
Utilization mark (high/low) : 100 / 0
Subnet size (first/next)    : 0 / 0
Total addresses              : 62
Leased addresses             : 1
Pending event                : none
1 subnet is currently in the pool :
Current index      IP address range      Leased addresses
129.70.1.42        129.70.1.1 - 129.70.1.62              1

Pool NT 2 :
Utilization mark (high/low) : 100 / 0
Subnet size (first/next)    : 0 / 0
Total addresses              : 254
Leased addresses             : 1
Pending event                : none
1 subnet is currently in the pool :
Current index      IP address range      Leased addresses
129.70.0.23        129.70.0.1 - 129.70.0.254    1

Router#
Router#
Router#
Router#
Router# ip dhcp binding
Bindings from all pools not associated with VRF:
IP address          Client-ID/      Lease expiration   Type
Hardware address/   User name
50.0.0.2             0150.190d.06ae.00 Aug 05 2026 08:17 AM Automatic
50.0.1.2             0150.b35e.06ad.00 Aug 05 2026 08:14 AM Automatic
50.0.1.130           0150.e9a6.06ac.00 Aug 05 2026 08:10 AM Automatic
129.70.0.53          0150.1641.06b6.00 Aug 05 2026 07:02 AM Automatic
129.70.1.41          0150.09b3.06af.00 Aug 05 2026 07:04 AM Automatic
Router#
```

IP auto Assigned in PC

